

behind the glove box or in the engine compartment.

An ECM manages the key engine operations, such as spark timing, fuel delivery, emissions and, in some cases, the automatic transmission too. The computer gets electrical signals from sensors and input devices associated with the engine on a regular basis. It analyses this information and sends control signals to valves, controllers and other output devices, to balance the requirements of power, fuel economy and emission control.

The ECM uses closed-loop control, a control method that monitors outputs of a system to control the inputs to a system, managing the emissions and fuel economy of the engine (as well as a number of other variables). Collecting data from a number of different sensors, the ECM keeps track of everything from the coolant temperature to the quantity of oxygen in the exhaust. Using this data, it performs millions of calculations per second, which include looking up values in tables, calculating the results of long equations to decide on the optimal spark timing and determining how long the fuel injector is open. The ECM works with the objective of achieving the lowest emission levels and highest mileage.

A contemporary ECM could even have a 64-bit, 100MHz processor. While this processing power may seem insignificant (considering the levels to which today's computers have reached), actually your car computer is more efficient than your PC. The programming memory needed by a typical ECM is approximately 1MB to 2MB. Whereas, generally, we need about 2GB of programming space on our PC. The microprocessor is packaged with several other components on a multi-layer circuit board. Besides, there are several other components in the ECM that support the processor. Some of these are mentioned below.

Analogue-to-digital converters. These read the outputs of certain sensors in the car. The output of a sensor is an analogue voltage. Since the processor understands only digital numbers,

the analogue-to-digital converter transforms this voltage into a 10-bit digital number.

Digital-to-analogue converters. Sometimes the ECM has to give an analogue voltage output to certain engine components. As the processor on the ECM is a digital device, it requires an element that can convert the digital number into an analogue voltage.

High-level digital outputs. In some of the recently launched cars, the ECM fires the spark plugs, opens and closes the fuel injectors and switches the cooling fan on and off. Such computerised tasks need digital outputs. For example, an output for controlling the cooling fan might supply 12V and 0.5A to the fan relay when it is on, and 0V when it is off. The minute amount of power that the processor supplies energises the transistor in the digital output, allowing it to supply a considerably higher amount of power to the cooling fan relay, which in turn provides an even higher amount of power to the cooling fan.

Communication chips. These chips carry out various communication standards that are used on cars. There are several standards used, but the most popular one is called controller-area networking (CAN).

Car computer doctor

Data streams, that is, the signals generated by on-board computers, constantly flow through operating system of the car, adjusting and re-adjusting the engine. Diagnostic computers that

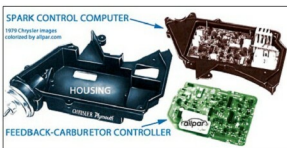


Fig. 2: Early automotive computer systems used in 1979 Chrysler cars (Source: www.alpar.com)



Fig. 3: ECM from a 1996 Chevrolet Beretta (Source: http://en.wikipedia.org/wiki/Engine_control_unit)



Fig. 4: A master diagnostic technician using a laptop computer to diagnose and repair the brake system (Source: <http://www.cbsnews.com>)

are interfaced with the car's computer read the data streams flowing through the system. If there is a problem with engine oil or car temperature, the information is communicated by the car computer. The problem could be related to faulty electrical or mechanical component, damaged wiring, etc.

The car computer system is programmed to send a problem signal whenever required. This signal gets stored in the car computer's memory